CODE DIGEST
All your tech in one place

sfi
CET

ACADEMIC SUPPORT CELL
An Initiative by SFI CET Unit

FULL–STACK DEVELOPER

PHASE 01

Internet & Web Fundamentals

Concepts

- How the internet works
- Client-server architecture
- HTTP/HTTPS
- DNS
- Browsers
- APIs
- JSON

Technologies To Familiarize Yourself With

- Chrome DevTools
- Postman
- Insomnia
- cURL

Build

- Static personal website
- Landing page

PHASE 02

Frontend Foundations

Technologies & Libraries

- Bootstrap
- Tailwind CSS
- Sass
- Vite

Learn

- HTML
- CSS
- JavaScript
- Responsive Design
- Accessibility

Familiarize Yourself With

- Shadcn UI
- Material UI
- Chakra UI
- Ant Design

Build

- Portfolio website
- Weather app
- Expense tracker

PHASE 03

Version Control & Developer Workflow

Learn

- Git basics
- Branching
- Pull Requests
- Merge conflicts

Technologies

- Git
- GitHub
- GitLab
- Bitbucket

Familiarize Yourself With

- GitHub Actions
- Conventional Commits

Build

- Maintain all projects in GitHub

PHASE 04

Frontend Frameworks

Technologies & Libraries

- Bootstrap
- Tailwind CSS
- Sass
- Vite

Learn One Deeply

- React
- Vue
- Angular
- Svelte

Familiarize Yourself With

- React Router
- Redux Toolkit
- TanStack Query
- React Hook Form

Build

- E-commerce frontend
- Blog frontend
- Dashboard

PHASE 05

Backend Development

Learn

- REST APIs
- Authentication
- Authorization
- JWT
- Middleware

Technologies

- Node.js
- Express
- FastAPI
- Spring Boot

Familiarize Yourself With

- GraphQL
- WebSockets
- gRPC
- Message Queues

Build

- Notes API
- Auth Service
- URL Shortener

PHASE 06

Databases

Technologies

- PostgreSQL
- MySQL
- MongoDB
- Redis

Learn

- SQL
- NoSQL
- Indexing
- Transactions

Familiarize Yourself With

- Prisma
- Drizzle
- SQLAlchemy
- TypeORM

Build

- Blog Backend
- Inventory System

PHASE 07

Authentication & Security

Learn

- Sessions
- JWT
- OAuth
- CORS

Technologies

- Auth.js
- Clerk
- Firebase Auth
- Supabase Auth

Familiarize Yourself With

- OWASP Top 10
- SQL Injection
- XSS
- CSRF

Build

- Secure Auth - System

PHASE 08

Full-Stack Frameworks

Technologies

- Next.js
- Remix
- Supabase
- Firebase

Learn

- SSR
- SSG
- Server Actions
- Full-stack Architecture

Familiarize Yourself With

- Server Components
- ISR
- Edge Functions
- Realtime Apps

Build

- SaaS Application

PHASE 09

Testing

Learn

- Unit Testing
- Integration Testing
- E2E Testing
- TDD

Technologies

- Vitest
- Jest
- Playwright
- Cypress

Familiarize Yourself With

- Mocking
- Coverage
- CI Testing
- Performance Testing

Build

- Add Tests To Existing Projects

PHASE 10

Deployment & DevOps

Technologies

- Vercel
- Render
- Docker
- GitHub Actions

Learn

- Hosting
- Domains
- CI/CD
- Containers

Familiarize Yourself With

- Kubernetes
- AWS ECS
- Nginx
- Reverse Proxy

Build

- Deploy All Projects

PHASE 11

Cloud Fundamentals

Learn

- Compute
- Storage
- Databases
- Monitoring

Technologies

- AWS
- Azure
- Google Cloud
- CloudFront

Familiarize Yourself With

- IAM
- VPC
- Load Balancers
- CloudWatch

Build

- Host Full Application